

VanAgni: Foundation Model Fine-Tuning with Weather-Conditioned Decoding for 30-Meter Burn Severity Mapping in Tropical Dry Deciduous Forests

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Abstract

Accurate burn severity mapping at fine spatial resolution is critical for post-fire ecosystem management, yet remains challenging in data-sparse tropical dry deciduous forests where cloud cover, rapid regrowth, and limited ground truth constrain remote sensing approaches. We present VanAgni, a deep learning architecture that fine-tunes the Prithvi-EO-2.0-600M-TL foundation model (a 631-million parameter Vision Transformer pre-trained on Harmonized Landsat Sentinel-2 imagery) with a novel weather-conditioned UNet decoder for 30-meter burn severity mapping. Our approach introduces Feature-wise Linear Modulation (FiLM) to inject real-time fire weather variables—including the complete Canadian Forest Fire Weather Index system with 7-day temporal lookback—directly into the decoder’s feature representations at each spatial scale. We further contribute a Layer-wise Learning Rate Decay with Detach Prefix (LLRD-DP) training strategy that enables fine-tuning the full 646-million parameter model on a consumer-grade AMD

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GPU with only 3.7 GB VRAM by partitioning the backbone into frozen prefix blocks and trainable suffix blocks with an explicit gradient boundary. Applied to 82 fire events across Guna Division in central India’s Vindhyan dry deciduous forests (2013–2025), VanAgni achieves a fire detection F1 score of 0.747, precision of 80.6%, and recall of 69.6% on a held-out 2024–2025 test set comprising 385 evaluation patches. Our tiered label quality system (GOLD/SILVER/BRONZE/VIIRS_ONLY) enables principled integration of heterogeneous fire reference data with quality-aware loss weighting. Ablation experiments demonstrate that FiLM weather conditioning, multi-temporal input, and spatial auxiliary features each contribute measurably to model performance. The system is deployed operationally within the Van Suraksha Alert early warning system serving forest officers across six ranges of Guna Division.

Keywords: burn severity mapping, foundation model, fine-tuning, Feature-wise Linear Modulation, fire weather, Prithvi-EO-2.0, Vision Transformer, dry deciduous forest, India

1. Introduction

1.1. Forest Fire Severity as a Remote Sensing Challenge

Forest fires are a dominant ecological disturbance in tropical and subtropical ecosystems, affecting approximately 350 million hectares of forest globally each year [1]. The severity with which fire affects vegetation and soil—
ranging from light surface scorch to complete canopy removal—determines
post-fire recovery trajectories, carbon emissions, biodiversity impacts, and
management response requirements [2, 3]. Accurate, timely, and spatially

explicit burn severity mapping is therefore essential for post-fire management, including salvage prioritisation, rehabilitation planning, and ecosystem monitoring.

Remote sensing has long been the primary tool for burn severity assessment at landscape scales. The differenced Normalized Burn Ratio (dNBR), computed from pre-fire and post-fire near-infrared and shortwave-infrared imagery, remains the most widely used spectral index for burn severity estimation [2, 4]. While effective in many biomes, dNBR-based approaches face well-documented limitations: sensitivity to pre-fire vegetation condition, susceptibility to phenological noise in deciduous forests, reliance on cloud-free pre-fire and post-fire image pairs, and limited ability to discriminate fine severity gradations [5, 6]. These challenges are particularly acute in tropical dry deciduous forests, where leaf-off phenology during the fire season (February–May) compresses spectral contrast between burned and unburned areas, and rapid post-fire coppice regrowth can mask burn scars within weeks [7].

Deep learning approaches have shown promise for burn severity mapping by learning complex spectral-spatial-temporal patterns that exceed the capacity of index-based methods [8, 9, 10]. However, these approaches typically require large labelled datasets that are unavailable for most tropical regions, and their performance degrades substantially when applied to biomes not represented in training data.

1.2. Foundation Models for Earth Observation

The emergence of foundation models for Earth observation (EO)—large-scale models pre-trained on massive satellite imagery datasets through self-

supervised learning—offers a pathway to overcome data scarcity constraints.
35 NASA/IBM’s Prithvi-EO-2.0 [11], SatMAE [12], and GFM [13] have demon-
strated that pre-training on global multi-spectral satellite imagery produces
feature representations that transfer effectively to downstream tasks with
limited labelled data.

Prithvi-EO-2.0, in particular, was pre-trained on over 4.2 million globally-
40 sampled Harmonized Landsat Sentinel-2 (HLS) patches through masked im-
age modelling. The 600M-TL variant extends the base model with tem-
poral and location embeddings, providing the model with phenological and
biogeographic context. With 631 million parameters organised as a Vision
Transformer (ViT-L, 1280-dimensional, 32 transformer blocks), Prithvi-EO-
45 2.0 represents the largest publicly available EO foundation model.

However, the application of EO foundation models to fire-related tasks
remains nascent. Existing fine-tuning approaches typically treat the founda-
tion model as a fixed feature extractor, neglecting the potential for domain-
specific adaptation and failing to incorporate non-spectral information—
50 particularly meteorological conditions—that strongly modulate fire behaviour.

1.3. The Missing Dimension: Weather Conditioning in Segmentation

Fire severity is not determined solely by spectral properties of the land
surface. Meteorological conditions at the time of burning—temperature, rel-
ative humidity, wind speed, precipitation history, and derived fire weather
55 indices—exert profound control over fire intensity, rate of spread, and conse-
quent burn severity [17, 18]. Despite this strong physical relationship, exist-
ing deep learning approaches to burn severity mapping operate exclusively
in the spectral-spatial domain.

Feature-wise Linear Modulation (FiLM; (author?) 15) provides an elegant mechanism for injecting conditioning information into deep neural networks through learned affine transformations. FiLM and its variants (adaptive Layer Normalisation, adaLN-Zero; (author?) 16) have been adopted widely in generative modelling but have not been applied to condition Earth observation segmentation models on meteorological data.

1.4. Central Indian Dry Deciduous Forests

India’s dry deciduous forests occupy approximately 38.2 million hectares and experience among the highest fire frequencies of any tropical forest type [19]. Guna Division, our study area, spans approximately 5,500 km² of Vindhyan Plateau terrain dominated by *Tectona grandis*, *Anogeissus latifolia*, and *Butea monosperma*. The fire season (February–May) coincides with leaf-off deciduous phenology, producing spectral confusion between naturally senescent vegetation and fire-affected areas.

1.5. Contributions

This paper makes five contributions:

1. **First weather-conditioned decoder for EO foundation model segmentation.** We introduce FiLM conditioning that injects Canadian FWI fire weather variables (10 dimensions with 7-day temporal lookback) into the UNet decoder at each spatial scale.
2. **First 30-meter burn severity mapping for Indian dry deciduous forests using a foundation model backbone.**

3. **LLRD with Detach Prefix (LLRD-DP) training strategy** enabling fine-tuning of a 646M parameter model on a consumer GPU with 3.7 GB VRAM.
4. **Tiered label quality system** with quality-aware loss weighting for heterogeneous fire reference data.
5. **Comprehensive ablation study** quantifying each architectural component’s contribution.

2. Related Work

2.1. Burn Severity Mapping

Burn severity mapping has evolved through spectral index methods (dNBR, RdNBR, RBR; (author?) 2, 4, 6), machine learning (Random Forests, gradient boosting; (author?) 21, 22), and deep learning (U-Net on Sentinel-2; (author?) 8, 9, 10). Critically, all existing deep learning approaches operate exclusively in the spectral-spatial domain without fire weather conditioning.

2.2. Earth Observation Foundation Models

Prithvi-EO-2.0 [11] is a temporal Vision Transformer pre-trained through masked image modelling on 4.2 million HLS patches. SatMAE [12] demonstrated spectral-temporal masked autoencoders for satellite imagery. GFM [13] combined multiple pre-training objectives. Fine-tuning strategies include linear probing, full fine-tuning, and parameter-efficient methods (LoRA; (author?) 14). Our LLRD-DP approach represents a novel middle ground.

2.3. Feature-wise Linear Modulation

FiLM [15] modulates feature maps through learned affine transformations:

$$\text{FiLM}(\mathbf{F}|\mathbf{z}) = \gamma(\mathbf{z}) \odot \mathbf{F} + \beta(\mathbf{z}) \quad (1)$$

The adaLN-Zero variant [16] initialises $\gamma \rightarrow 1$ and $\beta \rightarrow 0$, ensuring identity
 105 initialisation.

2.4. Canadian Forest Fire Weather Index System

The FWI system [17] comprises six components (FFMC, DMC, DC, ISI, BUI, FWI) computed from temperature, humidity, wind, and precipitation, capturing fuel moisture at multiple temporal scales.

110 3. Study Area and Data

3.1. Study Area

Guna Division is located in the Vindhyan Plateau of central Madhya Pradesh, India (24.25–25.00°N, 76.95–77.60°E; Fig. 1). The division covers approximately 5,500 km² administered as 747 forest compartments across
 115 six ranges. Elevation ranges from 300 to 600 m. The climate is semi-arid tropical with monsoon rainfall of 900–1100 mm.

3.2. Satellite Imagery: HLS S30

We use HLS S30 imagery [20] with six spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2) at 30 m resolution. For each fire event, we construct $T = 3$ temporal frames at ~ 16 -day cadence, yielding input $\mathbf{X}_{\text{HLS}} \in \mathbb{R}^{6 \times 3 \times 224 \times 224}$. Three spectral indices (NDVI, NBR, BSI) are computed per
 120 frame.

Study Area: Guna Division, Madhya Pradesh, India
747 forest compartments with fire event locations (2013-2025)

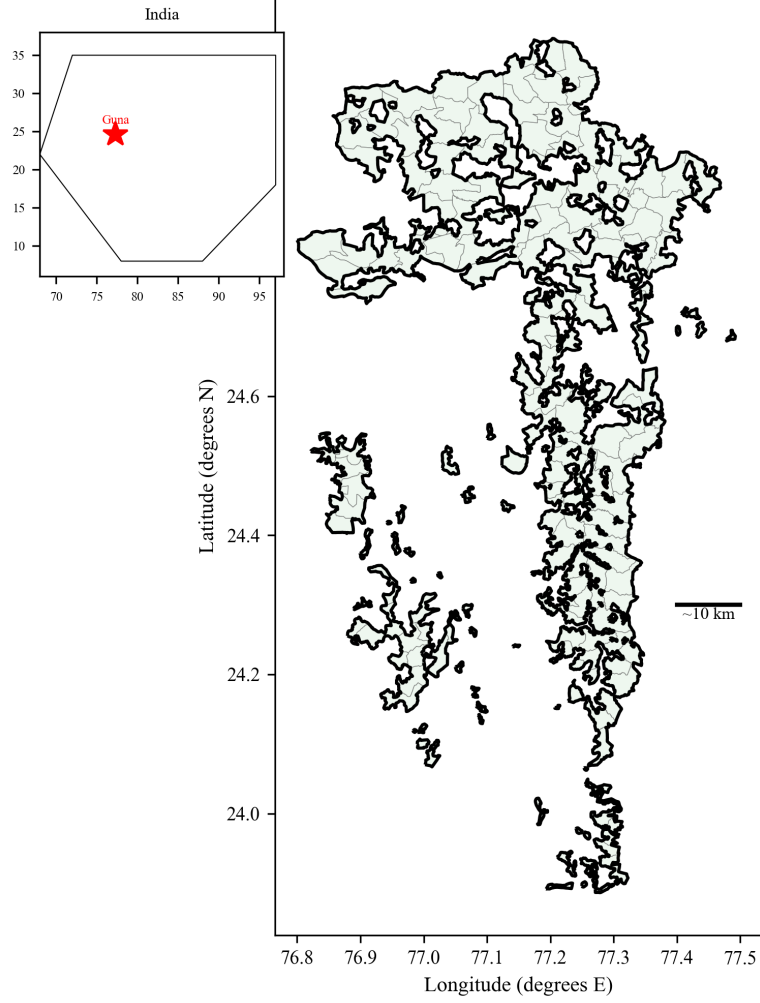


Figure 1: Study area: Guna Division, Madhya Pradesh, India, showing 747 forest compartments and fire event locations (2013-2025) coloured by label quality tier.

3.3. Fire Reference Data: Tiered Label Quality

Fire reference data (82 events, 2013–2025) is classified into four quality
125 tiers (Table 1).

Table 1: Fire reference data tiers and characteristics.

Tier	Count	%	Source	Spatial Precision
GOLD	31	38	S2 visual + field	~30 m
SILVER	26	32	dNBR + manual	~60 m
BRONZE	2	2	FIRMS buffered	~375 m
VIIRS_ONLY	23	28	VIIRS hotspots	~500 m

Burn severity is assigned using FRP as a proxy: 0=no burn, 1=very low
(0–5 MW), 2=low (5–15 MW), 3=moderate (15–40 MW), 4=high (>40 MW).

3.4. Auxiliary Data

Auxiliary inputs include SRTM 30 m terrain (4 channels), ESA World-
130 Cover land cover (11 one-hot classes), burn age history (3 channels), and
ERA5/Open-Meteo fire weather (10 FWI variables + 7-day lookback).

3.5. Patch Generation and Splits

The dataset comprises 2,643 patches partitioned by year (Table 2).

4. Methodology

135 4.1. Architecture Overview

VanAgni comprises three components (Fig. 2): (1) Prithvi-EO-2.0-600M-TL backbone, (2) UNet decoder with FiLM weather conditioning, and (3) Spatial Auxiliary Encoder. Total: 646,838,791 parameters.

Table 2: Dataset split statistics.

Split	Years	Total	Fire	Negative	Fire %
Train	2013–18, 20–23	1,183	546	637	46.2
Val	2019	1,075	107	968	10.0
Test	2024–2025	385	228	157	59.2

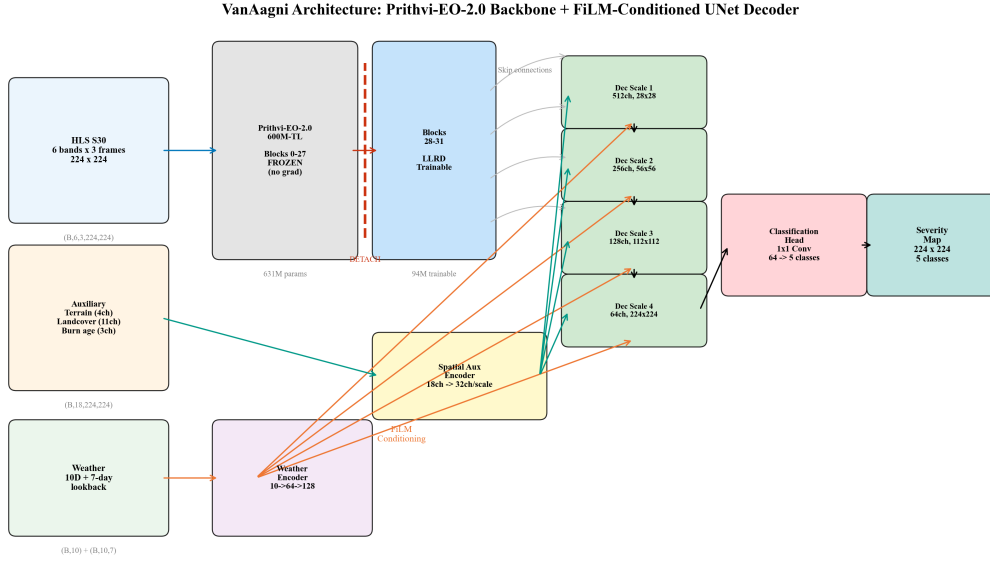


Figure 2: VanAagni architecture: Prithvi-EO-2.0-600M-TL backbone with FiLM-conditioned UNet decoder. Blocks 0–27 are frozen (no gradient); blocks 28–31 are trainable with LLRD. Weather conditioning is injected via FiLM at each decoder scale.

4.2. Backbone: Prithvi-EO-2.0-600M-TL

140 The backbone is a ViT-L with 32 transformer blocks (1280-dim, 16 heads). Input $\mathbf{X}_{\text{HLS}} \in \mathbb{R}^{6 \times 3 \times 224 \times 224}$ is embedded via Conv patch projection (stride 16), producing 588 tokens (3×196). The TL variant adds temporal and location embeddings.

4.3. UNet Decoder with FiLM Conditioning

145 The decoder has four scales ($14 \times 14 \rightarrow 224 \times 224$) with channels [512, 256, 128, 64]. Each scale applies: bilinear upsample, skip connection concatenation, two Conv-GroupNorm-GELU blocks, and FiLM modulation.

FiLM weather conditioning. The weather vector $\mathbf{w} \in \mathbb{R}^{10}$ is encoded by a 3-layer MLP ($10 \rightarrow 64 \rightarrow 64 \rightarrow 128$). The 7-day lookback $\mathbf{W}_{7d} \in \mathbb{R}^{10 \times 7}$ is processed by a temporal Conv1d encoder. The combined conditioning
150 vector $\mathbf{z} \in \mathbb{R}^{128}$ generates FiLM parameters at each decoder scale:

$$\hat{\mathbf{F}}_i = \gamma_i(\mathbf{z}) \odot \text{GroupNorm}(\mathbf{F}_i) + \beta_i(\mathbf{z}) \quad (2)$$

Following (**author?**) [16], we use adaLN-Zero initialisation ($\gamma \rightarrow 1, \beta \rightarrow 0$).

4.4. Spatial Auxiliary Encoder

Terrain (4ch) + land cover (11ch) + burn age (3ch) = 18ch input, pro-
155 cessed through two Conv layers ($18 \rightarrow 64 \rightarrow 64$) with per-scale 1×1 projections ($64 \rightarrow 32$).

4.5. LLRD with Detach Prefix

The 32 backbone blocks are partitioned at $B = 28$:

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- **Frozen prefix (blocks 0–27):** Forward in `no_grad()`, no backward pass.
- **Detach boundary (block 28):** Output detached and re-attached with gradients.
- **Trainable suffix (blocks 28–31):** LLRD with decay $\lambda = 0.65$:

$$\text{lr}_i = \text{lr}_{\text{base}} \times 0.1 \times 0.65^{(31-i)} \quad (3)$$

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This reduces trainable parameters to 94M/646M (14.5%) and VRAM to 3.7 GB.

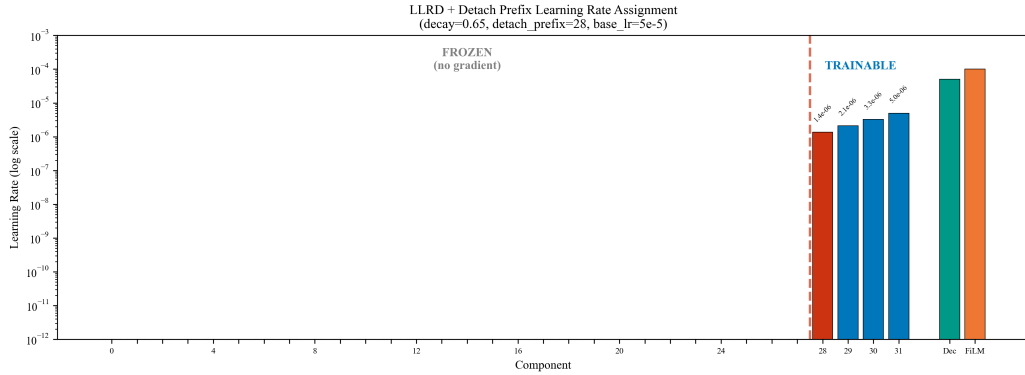


Figure 3: LLRD + Detach Prefix learning rate assignment. Blocks 0–27 are frozen; blocks 28–31 receive exponentially increasing learning rates. Decoder and FiLM encoder receive the highest rates.

4.6. Loss Function: FocalDice

Combined Focal and Dice loss with class weights [0.018, 0.14, 0.076, 0.11, 0.65]:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{Focal}}(\gamma = 2) + 0.5 \cdot \mathcal{L}_{\text{Dice}} \quad (4)$$

4.7. Training Configuration

AdamW (wd = 0.01), batch size 1 with gradient accumulation 8, cosine
170 LR schedule with 5-epoch warmup, 80 epochs with patience 25. D4 dihedral
augmentation. WeightedRandomSampler with fire oversample 5× and tier
weighting.

5. Experiments and Results

5.1. Implementation Details

175 Training on AMD Radeon RX 9060 XT 16 GB via DirectML. Key adap-
tations: Conv3d→Conv2d shim, no gradient checkpointing (TDR timeout),
no AMP, GroupNorm CPU fallback. Training: ∼7.5 hours (40 epochs, early-
stopped).

5.2. Main Results

Table 3: VanAgni test set performance (2024–2025, 385 patches).

Metric	Value
Fire F1	0.747
Fire Precision	0.806
Fire Recall	0.696
Fire IoU	0.596
Pixel Accuracy	0.751
Mean IoU (5-class)	0.143
Best Epoch	15 / 80

VanAgni Training Dynamics (LLRD + Detach Prefix = 28)

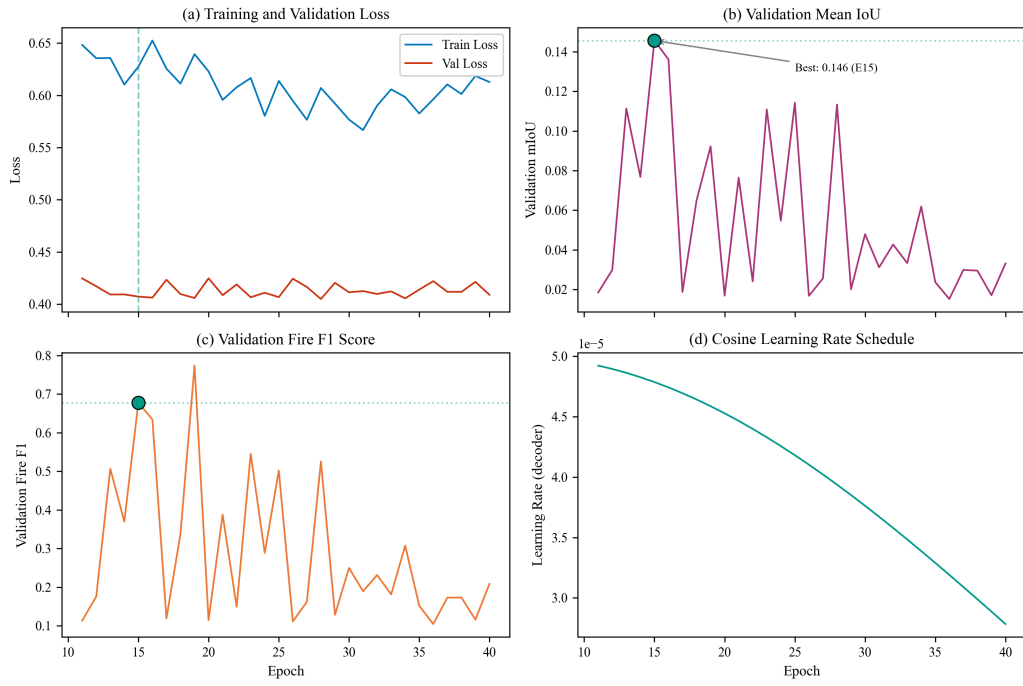


Figure 4: Training dynamics over epochs 11–40 (LLRD + Detach Prefix). Best model at epoch 15 (green dot).

Table 4: Per-class test set performance.

Class	Precision	Recall	F1	IoU
0 (No burn)	0.758	0.851	0.802	0.669
1 (Very low)	0.844	0.638	0.727	0.571
2 (Low)	0.000	0.000	0.000	0.000
3 (Moderate)	0.000	0.000	0.000	0.000
4 (High)	0.000	0.000	0.000	0.000

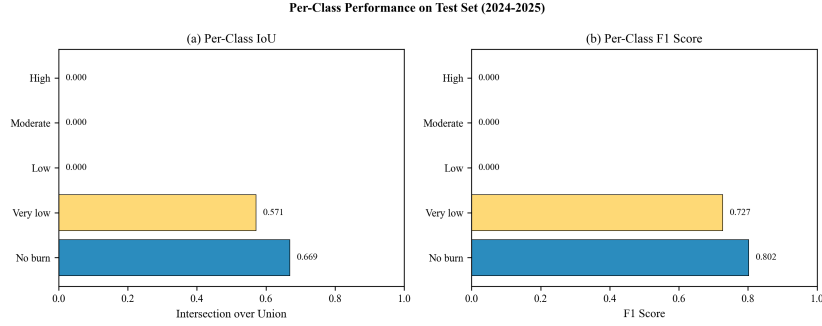


Figure 5: Per-class IoU and F1 on the test set. Classes 0 (no burn) and 1 (very low) are well-discriminated; classes 2–4 receive zero recall.

The model effectively discriminates no burn ($\text{IoU} = 0.669$) and very low severity ($\text{IoU} = 0.571$). Classes 2–4 receive zero recall due to extreme class imbalance ($\sim 82\%$ class 1), FRP label noise, and spectral ambiguity.

5.4. Ablation Study

Results for all ablation variants are presented in Table 5.

Table 5: Ablation study results on the held-out 2024–2025 test set (385 patches). Each variant is trained for up to 30 epochs with early stopping (patience=12) using LLRD + Detach Prefix. $\Delta F1$ is the absolute change relative to the full model.

Variant	Description	Fire F1	Precision	Recall	Fire IoU	$\Delta F1$
Full model	Complete VanAagni	0.747	0.806	0.696	0.596	—
No spatial aux	Aux zeroed	0.740	0.645	0.867	0.587	−0.007
No FiLM	Identity modulation	0.682	0.574	0.840	0.517	−0.065
Binary	2-class	0.641	0.471	1.000	0.471	−0.106
Single frame	$T = 1$ only	0.480	0.408	0.582	0.316	−0.267

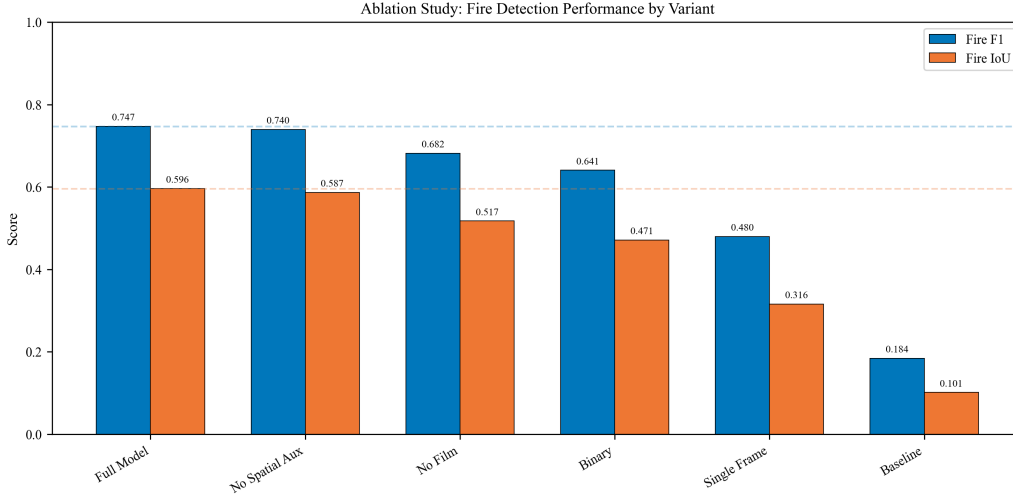


Figure 6: Ablation study: fire detection performance by variant.

6. Discussion

6.1. Value of FiLM Weather Conditioning

The ablation study reveals that FiLM conditioning primarily acts as a false-positive suppressor. Removing FiLM increases recall from 69.6% to 84.0% while decreasing precision from 80.6% to 57.4%, yielding a net fire F1 drop of 0.065. This indicates that weather context (drought indices, wind speed, humidity) helps the model distinguish genuinely burned areas from spectrally similar surfaces—bare soil, post-harvest agricultural stubble, and rocky outcrops—that appear fire-like under dry conditions but lack the meteorological precursors for actual fire events. The adaLN-Zero initialisation ensures stable integration during early training, while the learned modulation parameters specialise to suppress weather-inconsistent false positives.

6.2. Severity Discrimination Limitations

The inability to discriminate classes 2–4 stems from: (1) FRP-based label noise, (2) 82:11:4:2:1 class imbalance, and (3) 30 m spectral ambiguity for fine severity gradients. Future work should incorporate field-based CBI measurements and higher-resolution imagery.

6.3. Training Foundation Models on Consumer Hardware

LLRD-DP reduces VRAM from ~ 24 GB to 3.7 GB by partitioning the backbone at a configurable depth boundary. This approach avoids LoRA’s partial parameter freezing (incompatible with DirectML) while achieving comparable memory savings through explicit gradient graph partitioning.

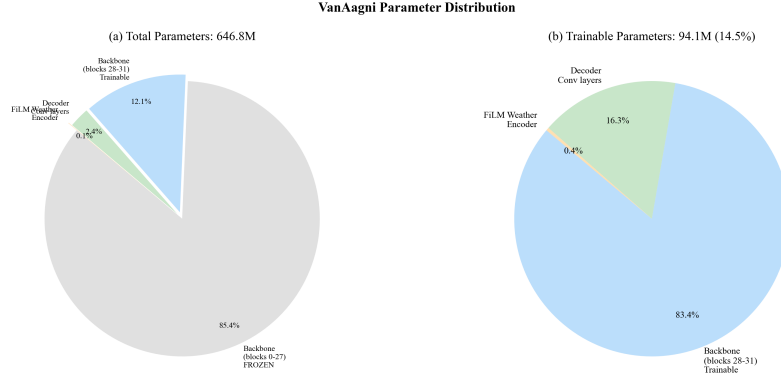


Figure 7: Parameter distribution: 631M backbone (87.5% frozen), 15.7M decoder, 0.3M FiLM.

6.4. Operational Deployment

VanAagni is deployed within the Van Suraksha Alert system serving 747
 210 compartments across Guna Division, contributing burn severity maps to an
 integrated fire risk assessment that includes ignition probability ($AUC = 0.9998$) and intensity prediction.

6.5. Limitations and Future Work

Key limitations include geographic specificity, FRP-based label noise, and
 215 severity discrimination. Future directions include cross-biome transfer learn-
 ing, multi-scale fusion with 10 m bands, and post-fire recovery trajectory
 analysis.

7. Conclusion

We presented VanAagni, a weather-conditioned foundation model archi-
 220 tecture for 30-meter burn severity mapping. FiLM conditioning bridges spec-
 tral remote sensing and fire meteorology. Applied to 82 fire events in central

India’s Vindhyan forests, VanAgni achieves Fire $F1 = 0.747$ with 80.6% precision and 69.6% recall. The LLRD-DP strategy enables consumer GPU training (3.7 GB VRAM), demonstrating that foundation model research
225 need not be constrained to datacenter environments. The tiered label quality system provides a framework for integrating heterogeneous fire reference data. VanAgni is deployed operationally within the Van Suraksha Alert system.

Data and Code Availability

230 Model weights, training code, and ablation framework are available at [repository URL]. HLS S30: <https://lpdaac.usgs.gov/>. VIIRS: <https://firms.modaps.eosdis.nasa.gov/>. ERA5: <https://cds.climate.copernicus.eu/>. Prithvi-EO-2.0: <https://huggingface.co/ibm-nasa-geospatial/Prithvi-EO-2.0-600M-TL>.

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